

MANDAKH, Mongolia

WMO: 443480

Lat: 44.4014N

Lon: 108.2492E

Elev: 1307

StdP: 86.58

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-24.0	-30.9	0.2	-22.5	-28.6	0.3	-20.5	15.7	-15.1	14.1	-17.5	6.6	320	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	32.2	17.9	30.4	17.2	28.7	16.6	21.0	28.6	19.6	26.9	18.4	25.7	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	15.5	25.5	17.0	14.2	24.1	15.8	13.2	22.5	67.4	28.4	62.2	26.9	58.0	25.5	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.2	(b) 12.3	(c) 10.5	(e) -28.7	(f) 35.1	(g) 2.9	(h) 1.7	(i) -30.8	(j) 36.3	(k) -32.5	(l) 37.3	(m) -34.1	(n) 38.2	(o) -36.2	(p) 39.4		
(4) 14.2	(5) 12.3	(6) 10.5	(7) -28.8	(8) 21.7	(9) 2.9	(10) 2.6	(11) -30.9	(12) 23.6	(13) -32.6	(14) 25.1	(15) -34.2	(16) 26.6	(17) -36.3	(18) 28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-15.4	-11.5	-2.3	6.4	13.0	19.5	22.5	20.6	14.3	5.3	-5.7	-13.3		
	DBStd	14.10	4.93	5.87	5.96	5.41	5.01	3.74	3.07	3.48	4.53	5.13	5.64	4.70		
	HDD10.0	3299	787	602	382	133	27	0	0	0	16	159	470	723		
	HDD18.3	5340	1046	836	639	359	175	30	4	16	132	403	720	981		
	CDD10.0	1307	0	0	1	24	121	284	387	329	145	15	0	0		
	CDD18.3	306	0	0	0	0	11	64	132	87	11	0	0	0		
	CDH23.3	2839	0	0	0	7	153	634	1198	752	93	2	0	0		
	CDH26.7	880	0	0	0	1	31	197	411	232	8	0	0	0		
(14) Wind	WSAvg	5.1	5.4	5.2	5.5	5.8	6.1	4.8	4.3	4.3	4.5	4.8	4.9	5.3		
(15) Precipitation	PrecAvg	130	2	2	4	3	10	20	32	31	13	7	4	2		
	PrecMax	217	5	12	16	10	43	36	86	78	35	23	12	7		
	PrecMin	79	0	0	0	0	1	2	10	1	1	0	0	0		
	PrecStd	33	1	3	4	3	10	9	15	20	9	6	3	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	5.5	16.7	24.0	30.2	33.7	35.3	33.9	27.8	21.7	10.7	0.4		
		MCWB	-4.8	0.2	5.6	10.5	13.6	19.0	21.6	19.4	14.3	10.9	3.8	-3.3		
	2%	DB	-3.9	2.3	12.6	21.0	27.3	31.1	32.4	31.2	25.9	18.5	7.5	-2.4		
		MCWB	-7.2	-2.0	3.7	8.8	12.7	15.9	18.6	18.4	13.7	9.4	1.9	-5.2		
	5%	DB	-6.3	-0.2	9.9	18.2	25.0	28.9	30.6	29.2	24.1	15.8	4.7	-4.7		
		MCWB	-8.6	-3.8	2.4	7.4	11.7	15.6	17.7	17.3	13.1	7.7	0.2	-7.0		
	10%	DB	-8.4	-2.5	7.0	15.7	22.7	26.9	28.9	27.4	22.2	13.3	2.5	-6.6		
		MCWB	-10.1	-5.4	0.5	6.4	10.8	15.1	17.4	16.4	12.4	6.2	-1.4	-8.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.4	0.8	6.1	12.6	16.1	21.5	24.2	21.8	17.6	12.5	4.2	-2.9		
		MCDB	-0.6	4.9	14.7	21.7	26.2	30.9	33.8	30.0	24.9	20.4	10.1	0.3		
	2%	WB	-6.9	-1.8	4.2	9.8	14.1	19.0	21.4	20.4	15.7	10.1	2.3	-5.1		
		MCDB	-4.1	2.1	11.9	18.9	23.6	26.9	28.9	28.3	23.2	17.5	7.3	-2.6		
	5%	WB	-8.5	-3.6	2.6	8.0	12.7	17.8	20.1	18.9	14.3	8.1	0.5	-6.8		
		MCDB	-6.4	-0.3	9.7	17.2	22.3	25.5	27.0	26.8	21.5	14.5	4.7	-4.8		
	10%	WB	-10.0	-5.4	0.7	6.5	11.5	16.5	19.0	17.7	13.1	6.4	-1.0	-8.4		
		MCDB	-8.4	-2.6	6.9	15.2	21.4	24.3	25.8	25.2	20.3	13.0	2.6	-6.6		
(35) Mean Daily Temperature Range	MDBR		8.4	10.0	11.6	11.6	12.2	11.0	10.2	10.3	10.8	10.2	8.9	7.7		
	5% DB	MCDBR	10.9	12.9	14.4	14.9	14.9	13.2	12.3	12.0	12.9	12.8	11.2	9.7		
		MCWBR	8.7	9.6	9.0	8.7	7.7	6.2	5.1	5.6	6.4	7.6	7.8	7.5		
	5% WB	MCDBR	10.7	12.6	14.1	14.1	13.8	11.6	10.4	10.6	10.8	12.2	11.0	9.6		
		MCWBR	8.6	9.6	9.1	9.0	8.1	6.4	5.8	5.8	6.6	8.0	7.9	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.279	0.317	0.366	0.401	0.424	0.399	0.379	0.346	0.306	0.269	0.253		
	taud		2.368	2.307	2.232	2.132	2.078	2.101	2.240	2.273	2.311	2.365	2.385	2.359		
	Ebn at Noon		882	914	915	892	868	847	865	871	873	868	853	844		
	Edh at Noon		75	97	120	145	159	156	134	125	111	90	73	68		
(44) All-Sky Solar Radiation	RadAvg		2.19	3.26	4.95	6.19	7.07	6.96	6.63	6.12	5.13	3.80	2.52	1.87		
	RadStd		0.21	0.32	0.19	0.28	0.30	0.38	0.34	0.21	0.19	0.14	0.15	0.20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.33	N/A	N/A	N/A	-0.85	-0.82	N/A	N/A	N/A	N/A

Nomenclature: See separate page